

Shalem Panthagani

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Professional Summary

Software Engineer with 2.5 years of experience in backend development and AI-driven systems, specializing in Java, Python, SQL, and with exposure to Go. Skilled in debugging and optimizing scalable services, designing resilient APIs, and resolving complex production issues under tight deadlines. Built and tested RAG pipelines and AI-driven debugging assistants, highlighting the ability to translate cutting-edge LLM research into reliable, user-facing solutions. Strong foundation in data structures, algorithms, and cloud infrastructure, with a proven track record of delivering high-impact solutions in both enterprise and academic environments.

Skills

Languages: Java, C, JavaScript, SQL, PHP, Python, HTML/CSS

Backend: REST APIs, jBASE, JSP/Servlets, Spring Boot, GraphQL(familiar)

System Design & Architecture: Distributed systems, scalability and performance optimization, high availability, fault tolerance, and microservices/API architecture (including caching and load balancing).

Networking Fundamentals: TCP/IP, DNS, HTTP/HTTPS, Routing Concepts (BGP basics)

Testing: JUnit, Automation Scripting, QA Processes, Agile/Scrum

Databases: MySQL, MariaDB, Oracle, ChromaDB

Build/DevOps: Bitbucket, Git, CI/CD, Linux

Frameworks & Tools: Postman, Swagger/OpenAPI, Jira, RTC (Rational Team Concert), Power BI

AI & NLP: Retrieval-Augmented Generation (RAG), LLM Integration, Natural Language Processing (NLP)

Experience

Associate Product Engineer

Temenos AG

Chennai, India

Dec 2021 – Mar 2024

- Streamlined client workflows and improved user experience by delivering 10+ enhancements and resolving 150+ production defects under tight SLAs.
- Automated transformation of proprietary .d/.b records into JSON format, enabling higher developer productivity and system-wide data interoperability.
- Led SCM migration from RTC to Bitbucket across 45 repositories; codified branching/PR workflow and mentored 5+ engineers.
- Debugged and resolved critical API throughput issues in production, improving reliability and availability of distributed client services.
- Owned the full software development lifecycle under the One-Engineer model (from reproduction through testing and release), improving delivery speed and code quality.

Associate Software Engineer

MAQ Software

Hyderabad, India

Apr 2021 – Jul 2021

- Enhanced Power BI dashboards with new measures/KPIs, improving visualization usability by 20
- Optimized queries, reducing refresh times by 10–15% and improving usability.
- Worked in Agile sprints to gather requirements, test deliverables, and implement enhancements.

Projects

RAG-Based Support System | *Python, sentence-transformers, ChromaDB, Azure OpenAI*

- Designed and deployed an end-to-end RAG pipeline integrating MiniLM embeddings, token-aware chunking, and ChromaDB for knowledge retrieval.
- Achieved a 12% improvement in answer accuracy vs. baseline and reduced hallucinations, demonstrating scalable firmwide knowledge access. Implemented top-k retrieval and cosine scoring to optimize performance for enterprise-scale queries.

Socratic Debugging Assistant | *Python, LLaMA-3.2, Prompt Engineering*

- Developed an LLM-powered assistant using Socratic questioning to guide students in debugging Python code.
- Developed a taxonomy of bugs and evaluation harness, creating a reproducible framework for AI-driven user support in knowledge-intensive workflows.

Education

Colorado State University - Fort Collins, CO

Master of Computer Information Systems

Aug 2024 – Dec 2025

- Coursework:** Database Management Systems, Software Product & Process Evaluation, Project Management (IT), Application Software Infrastructure

KL University

Bachelor's in Computer Science

May 2017 – Apr 2021